Data Analyst Job Postings Analysis

Xinyu ‘Sylvie’ Mei (xinyumei), Jiajing Zhu (jiajingz)

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Takeaway

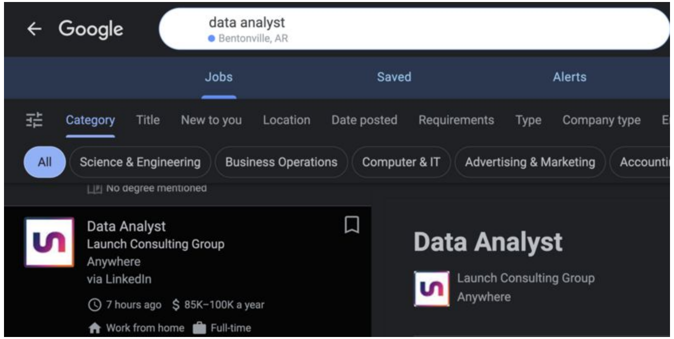
- **Binary Classification Performance**: **90%** accuracy (Random Forest and MLP)
- **Performance Degradation with More Categories**: As the number of salary categories increased (from 2 to 10), model accuracy declined significantly.

Number of Categories	Accuracy (%)
2	90
3,4	~80
5-7	Stable, ~75
8-10	<75

Question

- What factors impact salary difference? Can we **predict salary** based on job descriptions?
- What are the most **in-demand skills** across industries?
- Which **cities** have the highest **demand** for Data Scientists?
- What proportion of job postings are for **remote roles**? Which industries are leading in remote work adoption?
- What are the **seasonal patterns** in job postings? What is the basic trend in recent years? Specifically, what about Data Scientist entry level jobs?
- For those job postings without a clear label of **position level**, can we classify them to different levels (entry/junior/senior etc.) based on job postings?

Data



- **Data Analyst Job Postings (from Kaggle)**
  - Over 53,000 Data Analyst/Scientist job postings from Google search
  - First collected on Nov 4, 2022, with around 100 new postings added daily
  - Containing Data Analyst job postings information: job title, location, job description, salary, hard skills, posting date, etc.
  - Our main dataset

- **LinkedIn Job Postings 2023-2024 (from Kaggle)**
  - Over 124,000 job postings listed in 2023-2024 from LinkedIn
  - Including all kinds of positions (in which data positions are quite a small proportion)
  - Also including companies and industries information
  - Our supplementary dataset

We standardized salary data to an annual scale

Previous Methods and findings - EDA

- **Salary Distributions:**
  - Median salary: $86,000/year.
  - Salaries vary based on skills, location, and job level.
- **Remote Work Trends:**
  - **48.82%** of job postings are remote, reflecting the industry's shift toward flexibility.
- **Seasonal Hiring Patterns:**
  - Hiring peaks in late **summer** and **fall** (August–November).
  - Declining job postings over recent years, likely tied to broader economic trends.

Previous Methods and findings - EDA

- **Job Levels:**
  - Majority are senior-level roles (25,535 postings), followed by entry-level (3,280) and mid-level (2,095).
  - **Years of Experience**: Extracted for 24,478 postings, showing limited correlation with salary.
- **Dataset Limitations:**
  - Bias in location (MO, OK, KS, AR dominate).
  - Data sparsity with only 9,000 rows used after cleaning (41,000 original).

Methods and findings - Feature Engineering Refinement

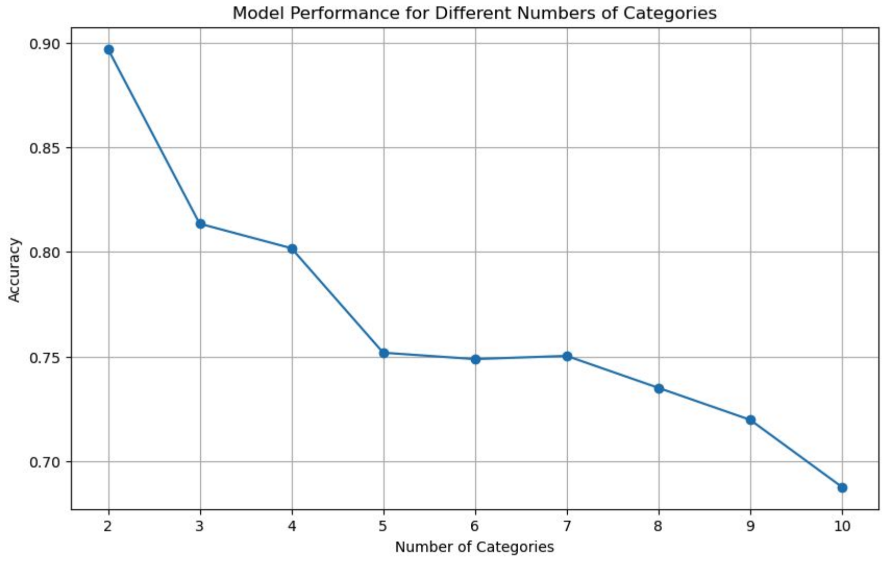
- Recapping last presentation: previous version of features got R-squared of 0.180 and MSE of 200+M for linear regression of salary prediction
- Deleted Job Description Length and Number of Required Technical Skills: 0 importance scores
- Industry encoding
  - Dropped the industry information table (389 industries in total) to ChatGPT-4o, asking it to summarize 21 parent categories to them, manually checked the quality
  - One-Hot encoded parent industries
- Location encoding
  - Only selected the states with top posting numbers (MO, OK, KS, and AR)
  - Categorized cities based on size (Top 6 in one state - major cities), combining minor cities in each state into one category, while major cities with their own categories (28 categories in total); then One-Hot encoding
- Job Description Embeddings
  - BERT
  - Word2Vec: utilized hard and soft skills as corpus
  - GLoVe: utilized glove.6B.100d.txt as corpus

Final Features Taking into Account

Feature Type	Feature Name	Description
Text-Based Features	Job level	Extracted by keyword matching, then encoding
	Year of Experience	Extracted by regex matching
	Hard Skills	Directly extracted by a column in the original df
	Soft Skills	Extracted by SkillNer
	Job Description TF-IDF	Extracted a TF-IDF matrix
	Job Description Embeddings	BERT, W2V (Hard & Soft skills as corpus), GLoVe
Categorical Features	Industry	Summarized parent industries, then encoding
	Location	Categorized into major, minor cities, then encoding
Numerical Features	Salary (Target)	Normalized to yearly amounts

Methods and findings - Modeling

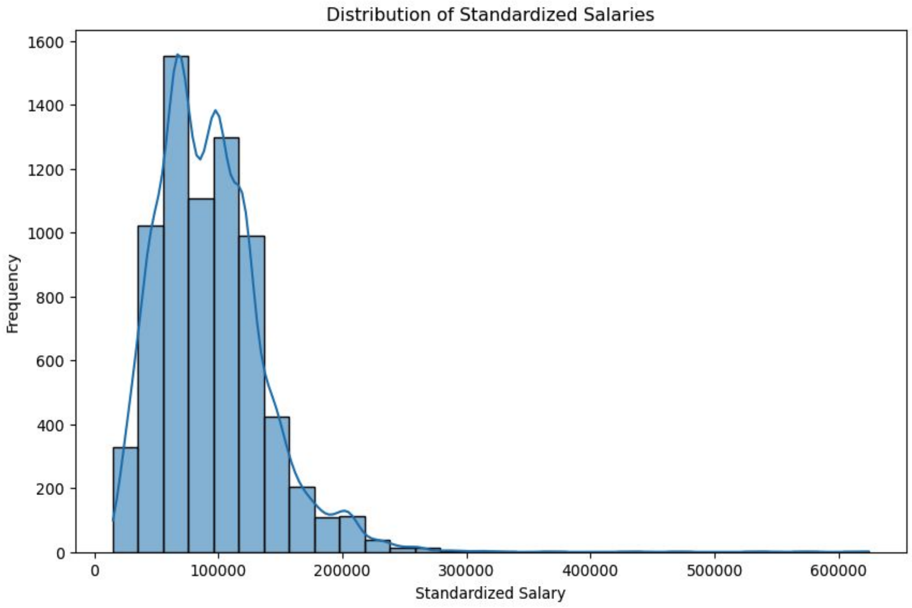
- Switched from regression to classification: starting from 2 categories (“high” & “low”)
- X = YOE + Job_Level + Industry + Location
- Fit salary~X + TF-IDF, salary~X + embeddings with different models
- The combination of X and W2V got best performance; the best models are Random Forest and MLP (ACC 0.9 in 2 categories classification)
- X + TF-IDF was ~0.01 worse than W2V in most models
- Extended to multi categories, accuracy dropped fast



Discussion - Key Findings and Explanations

- **Why classification overperforms regression?**
  - Simpler Goal
  - Reduced sensitivity to noise or uneven distribution in data
  - Evaluation Metrics

- **Why W2V or TF-IDF combining random forest and MLP got best results?**
  - Compared to GLoVe, W2V's corpus is more customized to this task
  - TF-IDF has comprehensive information about the documents
  - RF is robust to noise, while MLP gets large amount of information from the features



Limitations and Proposed Future Work

- **Limitations**
  - Data size might be insufficient
  - Location bias caused by the dataset publisher's location
  - Salary bins cut by qcut might not reflect real-world salary distinctions

- **Future Work**
  - Expand data collection and enhance diversity
  - Account for other external factors (education level, Glassdoor employee reviews, etc.)
  - Incorporate real-world salary distributions (from industry report, etc.)

References and Acknowledgements

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https://www.arkansas-demographics.com/cities_by_population

We got the idea of switching from regression tasks to classification ones from Justine’s feedback on 11-05 presentation